React Basics Notes

React is a JavaScript library for building user interfaces.

React applications are built using reusable components.

JSX allows developers to write HTML-like syntax inside JavaScript.

Props are used to pass data from a parent component to a child component.

State stores data that can change over time within a component.

The useState hook is used to manage state in functional components.

The useEffect hook is used for side effects such as API calls and timers.

React uses a Virtual DOM to efficiently update the user interface.

Keys help React identify list items when rendering arrays.

React Router enables navigation between pages in a single-page application.

Conditional rendering displays different UI based on conditions.

Event handling in React uses camelCase event names such as onClick.

Forms in React are commonly implemented using controlled components.

Context API helps share data without prop drilling.

Lifecycle behavior in functional components is handled using hooks.

Performance can be improved using React.memo and useMemo where appropriate.